1. Introduction and Use Case

Sleep is fundamental to physical and mental health. It is a significant determinant of the overall health, including cognitive ability, metabolic stability, and cardiovascular efficiency. Since sleep disorders have risen worldwide, digital research on health and clinical decision-making must include an in-depth study of the relationship between sleep quality and lifestyle factors.

The proposed project aims to provide a readily available interactive tool that can guide stakeholders in determining relationships among lifestyle and demographic variables and sleep outcomes. The main stakeholders will be students/researchers at the public health level (pattern discovery), doctors/health educators (risk factor communication, including sleep disorders), and self-management-interested individuals (understanding stress, activity, and sleep quality). The dashboard is designed to support responsible interpretation by articulating potentially confounding factors, in particular age, in the visual analytics.

This project uses the Sleep Health and Lifestyle dataset on Kaggle. It employs data-driven approaches to investigate the correlations among demographic, behavioral, and physiological variables and how they affect sleep quality and duration. The results are presented using an interactive R Shiny dashboard. With this tool, one can filter information, best represent essential trends, and gain a population-scale understanding of sleep health. The purpose is to demonstrate how interactive visual analytics can support digital health interventions, improve public health surveillance, and enhance preventive care activities.

2. Dataset Description

The dataset comprises 374 individuals with 13 variables, encompassing demographic information, lifestyle habits, vital signs, and sleep-related outcomes. The main variables include:

- Sleep Duration (hours): a numerical measure of nightly sleep
- Quality of Sleep (1–10): subjective sleep quality rating
- Stress Level (1–10): perceived stress
- Physical Activity Level (mins/week)
- Daily Steps: average daily walking steps
- Heart Rate (bpm)
- Blood Pressure (split into systolic/diastolic)
- BMI Category: Normal, Overweight, Obese
- Sleep Disorder: None, Sleep Apnea, Insomnia

The dataset is suitable for exploring associations between lifestyle behaviors and sleep outcomes, as well as for demonstrating the value of digital health dashboards in population-level analysis. In the given application, the dataset enables stakeholders to explore the sleep determinants in a self-directed manner and to support digital health communication by connecting lifestyle predictors (stress, steps, activity) with sleep outcomes and sleep disorder types.

3. Implementation and Solutions

3.1 Data Cleaning and Preparation

Data preparation was done in R using the tidyverse and janitor packages. Key steps included:

- Standardizing column names with clean_names().
- Converting categorical variables (gender, occupation, BMI category, sleep disorder) into factors.
- Splitting the blood pressure field ("126/83") into systolic and diastolic numeric variables.
- Ensuring correct numeric types for sleep duration, stress level, physical activity, daily steps, and heart rate.
- Removing rows with missing BMI categories only for the BMI-related visualizations.
- A cleaned dataset (sleep_clean.csv) was created for use in the dashboard.

3.2 Dashboard Development

The interactive dashboard was developed in R Shiny, combining reactive filtering with dynamic visualizations. Users can adjust:

- Age range
- Sleep disorder category
- BMI category

All visualizations update automatically based on these filters.

The solution was developed iteratively: (i) set of key relationships to be communicated (stress and sleep quality, BMI and sleep duration, activity and sleep quality), (ii) interactive filtering to help stakeholders explore, (iii) improvement of visualization idioms to prevent a misleading interpretation when discrete categories (sleep disorder) and demographic structure (age) are interacting.

The dashboard includes four analytical plots:

1. Sleep Quality vs. Stress Level: A scatter graph that portrays the negative relationship between sleep quality and stress. The respondents were grouped by age (18-29, 30-44, 45-59, and 60+) to facilitate age-conscious comparison.
   Additionally, the dashboard allows users to toggle between a scatterplot and a boxplot for stress-sleep analysis, which retains the discrete nature of the sleep disorder variable.
2. Sleep Duration by BMI Category: A boxplot comparing sleep duration across Normal, Overweight, and Obese groups.

3. Daily Steps vs. Sleep Quality: A plot visualizing how physical activity correlates with sleep quality.
4. Sleep Duration Distribution: A histogram revealing overall sleep duration patterns in the population.

The dashboard is hosted on shinyapps.io, making it accessible online without requiring local installation.

4. Results

4.1 Stress and Sleep Quality

Since sleep disorders are discrete variables and the prevalence of sleep apnea among the elderly population is clustered among older subjects, a single aggregated scatterplot may overstate prevalence and lead to unwarranted comparisons. Thus, the dashboard not only groups observations by age group (small multiples) but also provides a boxplot idiom to compare the distribution of sleep quality across sleep disorder categories in a more adequate way. The relationship between stress level and sleep quality was analyzed using age-aware visualizations to avoid misleading interpretations. Though a general scatterplot suggests a negative relationship between stress and sleep quality, this depiction masks the significant demographic structure of the data.

Figure 1a: Sleep quality versus stress level, faceted by age group.

When grouping participants by age, it is clear that sleep apnea cases are more pronounced in older age groups (45-59 and 60+). The fact that they can be plotted in the same scatterplot can therefore overstate the perceived high prevalence of sleep apnea in the entire population.

Figure 1b: Boxplot of sleep quality by sleep disorder, faceted by age group.

To overcome this shortcoming, the dashboard includes faceted scatterplots by age category and presents them in a different format: boxplots. Such visualizations help show that the stress-sleep relationship differs across age groups and that sleep apnea is not evenly distributed but is highly age-dependent.

4.2 BMI and Sleep Duration

The boxplot indicates that individuals classified as overweight have slightly shorter sleep duration, while the normal and obese groups show a wider range of values. After removing rows with missing BMI values, the comparison becomes clearer. Despite minor differences, the visualization suggests a relationship between metabolic factors and sleep habits.

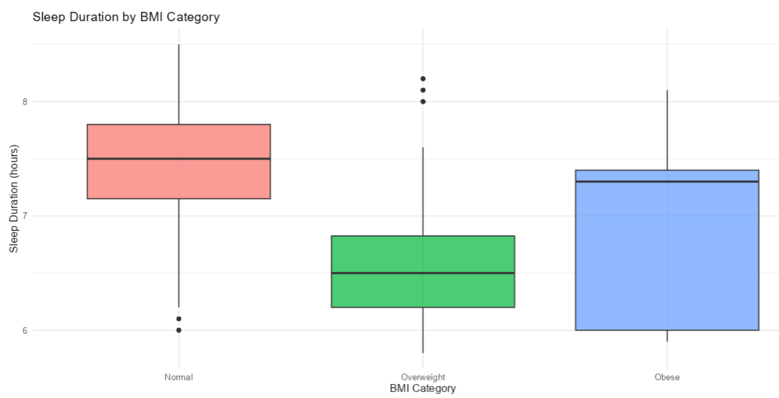

Figure 2: Distribution of sleep duration across BMI categories.

4.3 Physical Activity and Sleep

The level and the number of steps per day have a mild positive correlation with sleep quality. Those with more steps have higher inferred sleep scores, but the correlation is not linear. This implies that physical activity promotes healthy sleep.

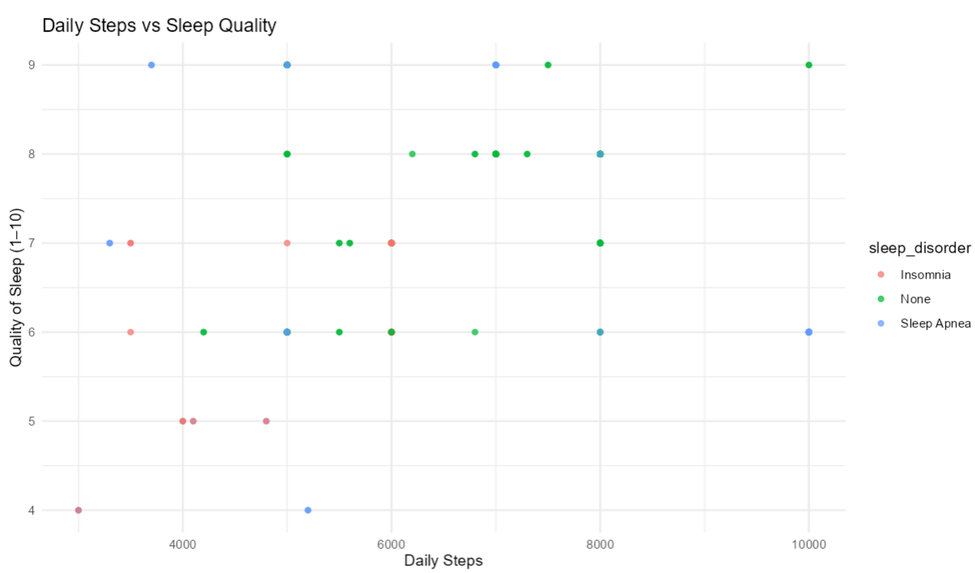

Figure 3: Association between daily steps and sleep quality.

4.4 Sleep Duration Distribution

The histogram shows that the individuals sleep 6-8 hours, with additional ranks below 6 hours and above 9 hours. This distribution is typical of adult sleep patterns and also identifies subgroups at risk of short sleep duration.

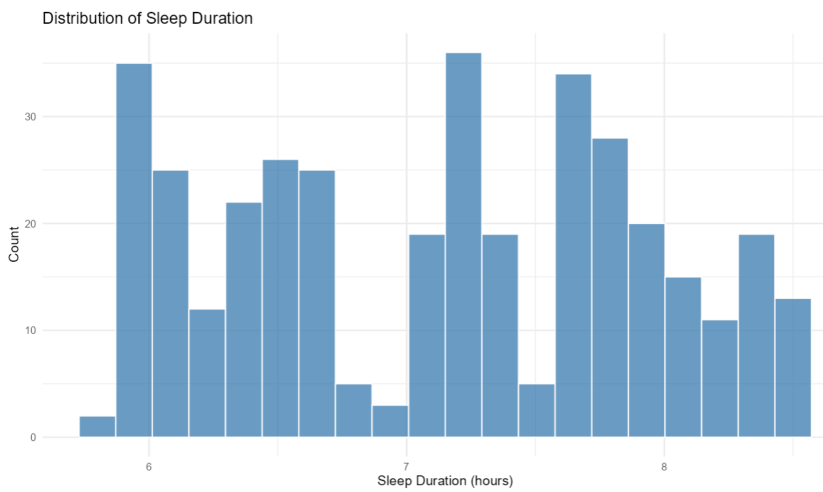

Figure 4: Distribution of sleep duration in the study population.

5. Implementations and Challenges

The dashboard is a convenient visualization tool for understanding how lifestyle and demographic factors interact to affect sleep outcomes. The most substantial effect is a negative relationship between stress and sleep quality, suggesting that sleep health can be improved through stress management interventions. The weak yet positive correlation between sleep quality and physical activity underscores the importance of exercise.

It was implemented in R Shiny using reactive filtering and modular plotting. The main issues were how to manage factor levels and missing values (e.g., BMI NA values) without distorting comparisons, guarantee uniform file paths so the app can be run locally and deployed, and determine which chart idioms to use that fit the data structure. These were handled by filtering out missing values where needed in the BMI-specific plot, by loading the relative project paths, and by introducing age-group faceting together with an alternative boxplot representation that compared discrete sleep disorders.

The data have limitations, including self-report (e.g., sleep quality, stress), limited demographic diversity, and a cross-sectional design. Also, Body composition differences are not necessarily represented by BMI categories. However, the dashboard indicates that interactive analytics can support the use of digital health applications and the development of data-driven health knowledge.

6. Conclusion

The given project demonstrates that combining data science and visualization, along with concepts of digital health, should be considered for analyzing sleep-related behaviors. The R Shiny dashboard offers an intuitive interface for examining lifestyle determinants of sleep health and can be expanded with predictive models or clinical decision-support features in future work.

7. Links

- **Shiny App:** https://kashfiaanika.shinyapps.io/digitalhealth_sleep/
- **GitHub Repository:** https://github.com/kashfiaanika/DigitalHealth-SleepHealth

Submitted by: Anika Tabassum Kashfia

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